# Lab 4 Writeup

Team 4 • Three-zone vehicle PCB • 29 October 2026

## 1. Design brief

We are designing a four-layer PCB that replaces the breadboards, two Nucleos and six CAN modules. We keep the Pi 4 and use two STM32F401RET6 MCUs for the front/rear zones. Only power, ground and dual CAN cross zone boundaries. All required PCB connections are routed; electrical power qualification and final fabrication review are unfinished. We have not powered the new PCB.

We retain the course L298N and motors. The handout's LD-1501MG servo has a published 6–8.4 V range; we propose a separate 6.6 V supply. Actual tolerance, ripple and wiring drop are unmeasured.

### Power budget

| Rail | Loads / proposed source | Peak / assumed average | Still to establish |
|---|---|---|---|
| Battery / motor | L298N; 3S LiPo, 11.1 V nominal / 12.6 V full | 6.4 A raw stall sum; avg 0.72 A planning case | Loaded voltage, driver drop, current limit, heat |
| 6.6 V servo | LD-1501MG; Separate buck | 4 A provisional; avg 0.39 A planning case | Idle, loaded and peak current; sag |
| 5.1 V | Pi + 5 V devices + encoder LDO; TPS564252 buck | 3.3 A allocation; avg 3.3 A assumed | Startup, cable drop, overshoot and heat |
| 3.3 V logic | MCUs, CAN and analog; encoders use a separate 4.0 V LDO | 0.8 A allocation; avg 0.8 A assumed | Load sum, ripple, heat and ADC reference |

Calculated planning case: two motors at catalog rated 0.36 A each; servo at 100 mA for 90% and 3 A for 10%; electronics continuously at their allocations. At 11.1 V and assumed 85% converter efficiency, Ibat = 0.72 + (6.6×0.39 + 5.1×3.3 + 3.318×0.8)/(11.1×0.85) = 3.058 A. At 10.5 V and 80% it is 3.346 A before input-path losses; our shared-track/FET planning model gives 3.41 A. Duty cycle, efficiency and the motor winding-current proxy are assumptions, not measured driving current or guaranteed bounds. Simultaneous motor stall is not a permitted L298 operating condition. Motor stall and L298 drop/temperature remain unmeasured.

### Battery choice

We selected the CNHL S220403BK shorty 3S LiPo (11.1 V, 2.2 Ah) and Hitec RDX1 Mini 44295 charger. The TA approved this pair on October 28; Amazon is permitted for both. CNHL advertises 40C continuous, equivalent to 88 A for 2.2 Ah; this is a supplier rating, not a measured or permitted vehicle current. The BOM requests one two-pack listing (one fitted, one spare) and one charger. Standard 3S LiPo balance charging uses 2.2 A and 4.20 V/cell, off the car with both leads connected and a charging bag. Nothing has been ordered by this revision.

Protected battery feeds the L298 and three TPS564252 bucks. The 8–15 V design screen is not a discharge cutoff. We added a 100k/10k/100nF divider on Front PA3 and a standalone guard proposing a latched stop at 10.5 V; MCU integration and demonstration are unfinished. A total-pack reading does not protect individual cells. The input includes a PMOS, 1000 µF bulk capacitor and a 10 A MINI fuse candidate; an added pack-side fuse holder protects the upstream harness. Fuse coordination is unqualified and does not limit each motor to 2 A.

## 2. Current-sensing design

Calculated sensor values assume a 12-bit ADC and 3.3 V reference. Ideal quantization is not measurement accuracy.

| Channel | Proposed range | ADC scale | Power / filter | Test points |
|---|---|---|---|---|
| Left motor | −5 to +5 A | 8.54 mA/count at 5.1 V | Hall conductor ≈30 mW at 5 A; fc ≈265 Hz | LEFT_RAW, LEFT_SCALED, LEFT_ADC, GND |

| | | | | |
|---|---|---|---|---|
| Right motor | −5 to +5 A | 8.54 mA/count at 5.1 V | Same as left | RIGHT_RAW, RIGHT_SCALED, RIGHT_ADC, GND |
| Servo | 0 to 5 A | 1.61 mA/count | 10 mΩ ≥1 W; 0.25 W at 5 A; fc ≈339 Hz | SERVO_RAW, SERVO_ADC, GND |

Each motor's ACS712 sits in one winding lead after the bridge. Its 10k/10k divider and 1k/100nF filter give VADC ≈ 1.275 + 0.09435I at a nominal 5.1 V sensor supply; fc = 1/[2π(5k+1k)100nF] ≈ 265 Hz. Offset/gain need calibration.

Servo sensing uses Vishay WSL2512R0100FEA (10 mΩ, 1 W, 1%) and INA180A2 gain 50: VOUT = 0.5I. At 5 A this gives 2.5 V, 50 mV shunt drop and 0.25 W. Use Kelvin sense traces. The 4.7k/100nF filter gives 339 Hz. The measurement range does not authorize 5 A servo operation.

**What is the idle current of your servo holding position with no load, and what is the smallest change in load you want to feel on the steering wheel? What shunt value and gain does that imply?**

Hiwonder's exact LD-1501MG table lists 100 mA no-load and 2.4–3 A stall, without test voltage or static-hold/motion detail. Thus holding current at 6.6 V is unmeasured. Our 10 mΩ/gain-50 design gives 50 mV (62 ideal counts) at 100 mA. We target 20 mA changes: 10 mV or 12.4 counts. At 3 A the shunt dissipates 0.09 W. Noise and calibration remain unverified.

**The servo draws a pulsed current at its internal PWM frequency. What does your ADC see if you sample without a filter, and what averaging do you do in software?**

Unfiltered samples depend on pulse timing and can alias. We propose 4 sample/s per channel and a 16-sample moving average, with adequate ADC acquisition time. This firmware is not yet implemented or validated.

**The ACS712 has a 2.5 V zero-current offset that drifts with temperature. How do you calibrate it at power-up when the motor is guaranteed to be stopped?**

Disable the bridge, keep wheels stationary, average each channel's zero-current samples and subtract that offset. Do not zero during motion or braking. Calibrate gain against a reference current and retain signed readings.

## 3. Circuit architecture and part choices

The schematic contains 300 components across five sheets. Local support includes MCU VCAP/reset/BOOT0/SWD, six CAN clocks, decoupling, test points, a separate encoder regulator and the battery divider. The BOM records exact MPNs, quantities, spares and sources; the principal choices are below.

| Part / choice | Qty | Reason / remaining check |
|---|---|---|
| TPS564252DRLR buck converter | 3 | Control voltage of three main power rails (6.6V, 5.1V, and 3.3V) |
| STM32F401RET6 | 2 | Same MCU as Nucleo; support circuits placed; PCB firmware port pending |
| MCP2515T-I/ST + SN65HVD231DR | 6 each | Course-listed; two independent interfaces per zone |
| ACS712ELCTR-05B-T | 2 | Bidirectional motor sensing; 5.1 V supply, ADC divider |
| INA180A2 | 1 | INA180A2IDBVR: DigiKey-listed addition |
| WSL2512R0100FEA — 10 mΩ, 1 W, 1% | 5 | One installed + four spares; Kelvin sense routing |
| L298N | 1 | Course driver; fuse/thermal review open |
| heatsink | 1 | 6396BG accepted; two soldered anchors and M3 tab clamp implemented; thermal operation untested |
| flyback diodes | 8 | SK520B; course-listed, checked polarity |
| 16 MHz crystals (found on parts list) | 6 | Independent clocks; 18 pF loads assume 3 pF stray |

| | | |
|---|---|---|
| 120 Ω termination (found on parts list) | 4 | Two switchable end terminators per bus |
| LEDs (found on parts list) | 16 used | 12 node indicators + four vehicle LEDs; spares in BOM |
| Battery pack<br>CNHL S220403BK; 3S 11.1 V / 2.2 Ah LiPo | 1 | One fitted + one spare; approved Amazon B0DQP7SKGV |
| Hitec RDX1 Mini 44295 | 1 | 3S LiPo balance/storage; approved Amazon B07ZDKT6BD |
| Hirose DF63-2S-3.96C link | 3 | XT60 battery + two motor adapters; spares in BOM |
| Crimp connectors /DF63A-1618SC link | 6 | 16–18 AWG crimp contacts; XT60 end is soldered |
| Hirose DF63M-2P-3.96DSA link | 3 | Battery J40, left J80, right J81 |

Two linear CAN backbones run at 500 kbit/s. Six MCP2515/SN65HVD231 pairs have independent 16 MHz crystals, CS/reset pull-ups and decoupling; RS is grounded. Each node/bus interface has two isolation jumpers that disconnect H and L. Four switchable 120 Ω end terminators stay on the bus side.

Changes from our earlier hardware: bare MCUs replace Nucleos; the Pi uses a bottom-facing ESQ-118-44-L-D socket covering physical Pi pins 3–38; servo sensing becomes high-side shunt sensing; motor Hall sensors remain. We retain Lab 3 SPI/CAN/timing/LED assignments except the unavailable LQFP64 PB11 marker moves to PC9. New ADCs use PA2/PA3; servo PWM uses PA9, motor PWM PA9/PB7. Pin-map.csv records the rest. Lab 1 Section 12 records this revision. Lab 3 actuator PWM used PB4; the port to PA9/PB7 is unfinished. No new CAN IDs or real feedback are implemented here.

October 24 cable evidence: servo black/red/white maps to J90 GND/sensed 6.6 V/PWM (pins 1/2/3). The reported old 5 V bench supply is not the new rail. The existing plug needs an adapter; its fit/current capability remain unverified.

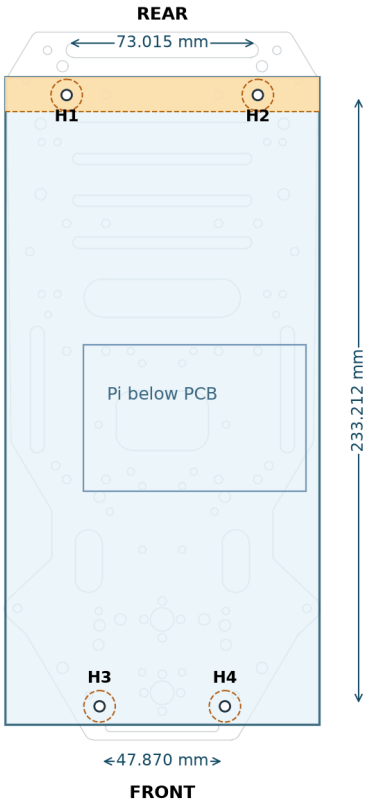
The photographed motor harness's jumper order is red motor-1, orange encoder supply, yellow A, blue B, gray encoder ground, black motor-2. Motor-2 is not ground. J80/J81 carry motor power; J82/J83 carry supply/GND/A/B. We added a 4.0 V MCP1700-4002E/TO supply with two TPSB106K016R0800 10 µF capacitors. The buffer and pull-ups stay at 3.3 V. At an assumed 20 mA per encoder, the conservative static screen gives 3.792–3.794 V after positive-feed loss, within the recorded 3.3–5 V range. This is calculated, not measured; adapter continuity remains an assembly check.

Buck feedback pairs are 100k/10k, 75k/10k (0.1%) and 45.3k/10k (0.1%) for 6.6/5.1/3.318 V, with 4.7 µH inductors. Revised 1210 input banks total 66 µF nominal; servo/Pi/logic outputs total 88/66/66 µF, with 68/91/30 pF feed-forward parts. The bias/tolerance/temperature/aging screen passes the stated capacitance targets; loop/transient and thermal performance are unmeasured. Our multilayer copper model gives 68 mV Pi-feed loss at 3 A. With assumed 80 °C copper, 10 mΩ contacts and 20 mV return loss, the screen gives 4.876 V before transients; full-load margin is unresolved. Each MCU uses 4.7 µF VCAP1.

**What are the advantages and disadvantages of high side current sensing? Low side current sensing? In line current sensing?**

High-side sensing preserves servo ground but needs supply common-mode tolerance. Low-side sensing raises load ground. In-line sensing captures bidirectional winding/braking current under switching common-mode voltage. We choose high-side shunt sensing for the servo and in-line Hall sensing for motors.

## Figure 1. Selected chassis supports and PCB dimensions




### Selected CAD pattern

PCB: 120 × 247.212 mm
Rear extension: 13.212 mm
Rear centers: 7 mm from edge

#### Chassis supports

M3 screws / 35 mm spacers
PCB drills: Ø4.0 mm, non-plated
Rear chassis holes: Ø3.2 mm in CAD
7 mm OD washers

Dashed circles: 6 mm radius
= 2 mm hole radius + 4 mm exclusion
Keep components and all copper out.

#### Team measurements

Front: approximately 48 mm
Rear A1/A2: 73.45 mm
Straight row spacing: 232.85 mm
Rear pair unused and accessible.

CAD nominal values selected by team.
Not a rigid-PCB fit test; drawing not 1:1.
Use Placement-fit-template.pdf at 100%.

## 4. Mechanical design, fabrication and verification

We selected the rear pair A1/A2, confirmed unused and accessible on our chassis. CAD dimensions are 47.870 mm front, 73.015 mm rear and 233.212 mm between rows; our checks measured about 48, 73.45 and 232.85 mm respectively. We extended the PCB rearward to 120×247.212 mm. Its 4.0 mm NPTH holes allow nominal 0.5 mm adjustment around M3 screws; we use 7 mm OD washers. Pi holes remain 2.7 mm. The 35 mm chassis/22 mm Pi stack is unchanged; the earlier cardboard fit used the previous pattern.

The October 24 cardboard check established a 115×50×50 mm battery envelope and 51.75 mm usable height. CNHL lists 77×33×26 mm; adding 5 mm to every dimension gives 82×38×31 mm, within that envelope. Pack retention and lead routing remain assembly checks. We specify 35 mm chassis and 22 mm Pi spacers, with Pi hardware at most 5.5 mm outside diameter. The shortened socket leaves 4.535 mm from the nearest Pi hole edge and 3.135 mm from that hardware. Actual mating and retention remain untested.

New purchases use DigiKey/Mouser, with Amazon permitted for the selected battery and charger. The purchasing CSV includes the shorter socket, encoder supply, precision resistors, battery monitor, revised capacitor banks and harness parts. Course allocations, the servo-side lead and fuse coordination remain flagged. U81 can use the SN74LVC14APW tube-pack version. The 6396BG has two soldered anchors and an M3 tab clamp from Boyd/ST drawings. We now use the CAD front pair and rear pair A1/A2, checked against our chassis measurements, with added PCB-hole allowance.

### Battery location

We place the battery low and near the center to reduce tipping. Completed-vehicle center of mass and drive stability remain unmeasured.

### Motor cable routing

The observed motor leads rise through the first and third thin chassis slots. We orient rear connectors near those exits. A mirrored chassis requires rechecking left/right connector location and cable reach before fabrication.

### Pi stack and access

Our Pi sits below the PCB, component-side up. The 36-contact socket covers Pi physical pins 3–38: align its first pair with Pi3/4, not1/2; pin4 alone supplies 5 V. The nominal 22 mm gap gives 5.17 mm insertion and 2.6 mm USB-body clearance after allowing 3 mm solder tails. The 35 mm chassis spacers leave 6.4 mm after the assumed Pi thickness and a 5 mm underside envelope. These are calculated clearances, not an assembled fit. We define plug/cooler envelopes and preserve access to top-side jumpers, buttons, SWD and probes.

Staff confirmed on October 26 that four layers are allowed. We use F.Cu / In1.Cu ground / In2.Cu signals / B.Cu, nominal 1.6 mm thickness, proposed 1 oz outer copper, 0.20 mm signal width/clearance, 0.50 mm copper-to-edge clearance and 0.30/0.70 mm via drill/pad. The saved JLC04161H-7628 stack specifies 1 oz outer / 0.5 oz inner copper; fabrication settings must match it. Buck footprints have a local 0.15 mm pad-gap exception. Chassis/Pi supports exclude components and copper within 4/3 mm of the finished hole edge on all layers. The socket clears the Pi mounting holes.

### CAN pair geometry

Target nominal CAN differential impedance is 120 Ω. Actual impedance depends on geometry/stackup; edge rate matters even at 500 kbit/s. We keep pairs together over a ground reference, with short stubs and A/B separation. Our routed four-layer board still needs final stackup and pair-geometry review; we do not claim controlled impedance.

ST limits L298 DC/braking current to 2 A per bridge. At 1 A each, its maximum 3.2 V bridge drop implies up to 6.4 W for two motors; at 2 A each, 4.9 V implies 19.6 W, before other losses. These are calculations, not temperature measurements. The TA confirmed on October 29 that extra hardware current limiting is optional. We keep the course circuit and omit the additional limiter and optional camera. We implemented the 6396BG's two solder anchors and screw-clamped L298 interface from manufacturer drawings. Brief-stall behavior and thermal performance are unverified.

We plan on team stencil assembly under the October 26 TA guidance. All 300 electrical footprints and the heatsink anchors are placed. Native ERC, DRC, schematic parity and connectivity report zero findings; 331 targeted connection checks pass. The battery guard passes host tests and ARM compilation but is not integrated or hardware-tested. Capacitor banks and chassis allowance are implemented. Full Pi supply margin, loop/thermal qualification, fuse coordination and manufacturing review remain open. This is not a fabrication release; no new PCB has been powered.

## Figure 2. Three-zone system architecture

Power is green; CAN A/B are blue/red. Current sensors lie in actuator current paths, with analog outputs to their owning MCU. The diagram source remains editable in Git.

